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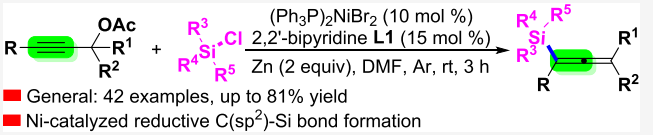
# Nickel-Catalyzed Reductive Cross-Coupling of Propargylic Acetates with Chloro(vinyl)silanes: Access to Silylallenes

Chong-Hui Xu, Zhi-Qiang Xiong, Jing-Hao Qin,* Xin-Hua Xu, and Jin-Heng Li*

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**ABSTRACT:** Because of their various reactivities, propargyl acetates are refined chemical intermediates that are extensively applied in pharmaceutical synthesis. Currently, reactions between propargyl acetates and chlorosilanes may be the most effective method for synthesizing silylallenes. Nevertheless, owing to the adaptability and selectivity of substrates, transition metal catalysis is difficult to achieve. Herein, nickel-catalyzed reductive cross-coupling reactions between propargyl acetates and substituted vinyl chlorosilanes for the synthesis of tetrasubstituted silylallenes are described. Therein, metallic zinc is a crucial reductant that effectively enables two electrophilic reagents to selectively construct C($sp^2$)−Si bonds. Additionally, a Ni-catalyzed reductive mechanism involving a radical process is proposed on the basis of deuteration-labeled experiments.




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## INTRODUCTION

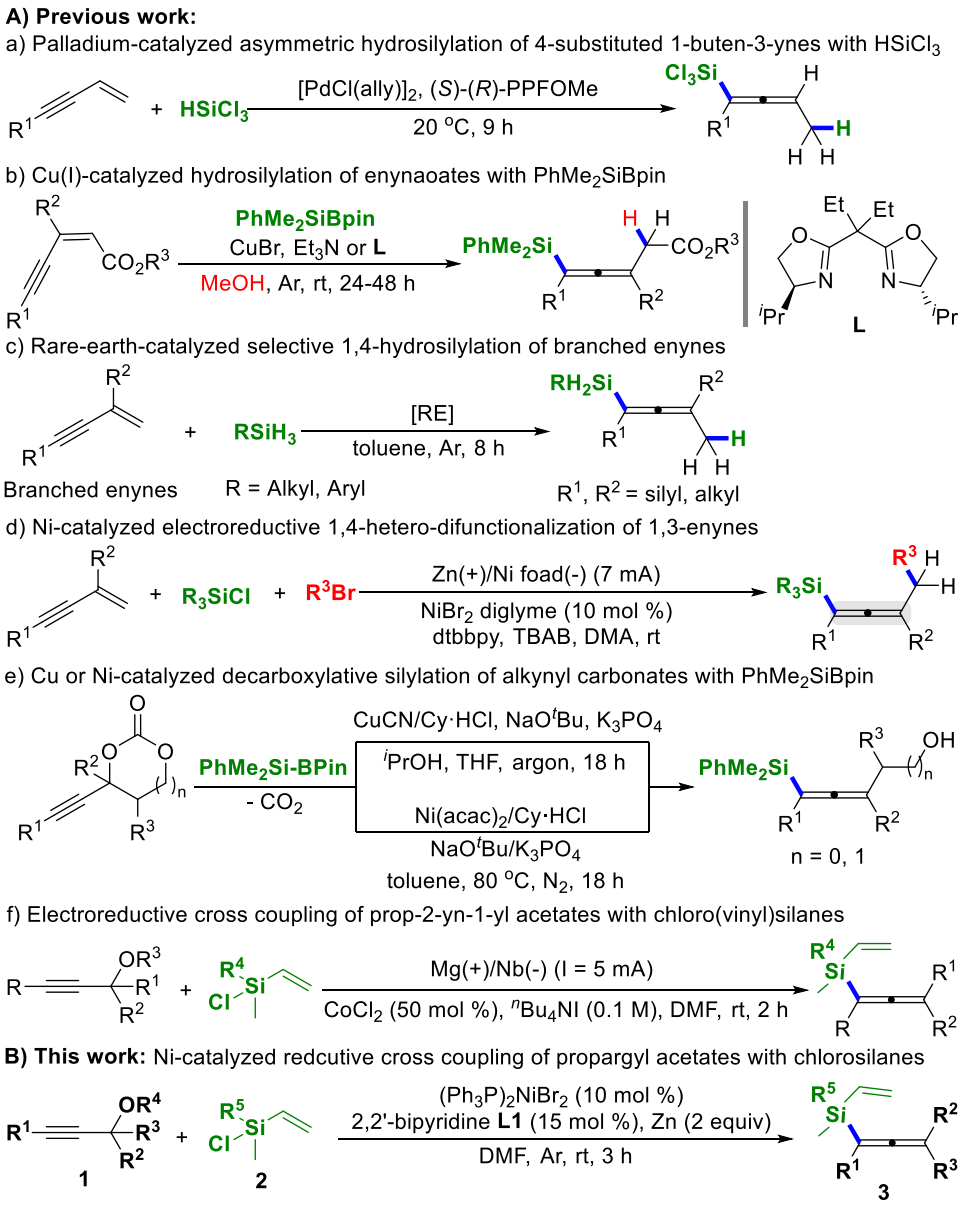
Organosilanes, such as silylallenes, are one of the most important functional structural units in fine chemicals, pharmaceuticals, and materials and generally emerged as versatile synthetic intermediates in chemical synthesis.[1] Accordingly, the development of efficient synthetic methodologies to access these functionalized organosilanes has drawn much ongoing interest. Among them, transition-metal-catalyzed C−Si bond construction reactions play an important role in such silyl compound preparation by incorporation of the silyl functionalities and have been well investigated in recent years.[1,2] Despite extensively documented methods for the C−Si bond formation,[2] reports of handling the transition-metal-catalyzed silylation of alkynes via the C($sp^2$)−Si bond construction to access silylallenes remain rare.[3,4] Pioneering work on transition-metal-catalyzed hydrosilylation of 1,3-diynes with trichlorosilane to prepare trichlorosilyl allenes has been reported by Hiyama and co-workers.[3] However, the method suffers from both a narrow scope of 1,3-diynes and limited resulting silylallenes. In 2001, Hayashi and co-workers extended a similar hydrosilylation strategy to realize the first palladium-catalyzed asymmetric hydrosilylation of 4-substituted 1-buten-3-ynes with $HSiCl_3$ for the synthesis of silylallenes (Scheme 1A(a)).[4a] However, only few available examples were examined. Subsequently, Loh, Xu and co-workers have reported a new efficient strategy for assembling silylallenes in particularly good yields that had high enantio- and regioselectivity based on the copper-catalyzed 1,6-silyl addition to enynoates with relatively expensive Si−B reagents (Scheme 1A(b)).[4b] Cui, Li and co-workers have employed the rare-element complexes for the effective catalytic 1,4-hydrosilylation of 1,3-enynes with hydrosilanes to construct tetrasubstituted silylallenes (Scheme 1A(c)), which used a wide range of 1,3-enyne substrates to greatly broaden the resulting silylallenes.[4c] Very recently, Rueping, Yue and co-workers have developed a new radical strategy to realize nickel-catalyzed three-component electroreductive cross-coupling for the 1,4-silylalkylation of 1,3-enynes via the activation of alkyl bromides to form the alkyl radicals, wherein chlorosilanes serve as the electrophiles to terminate the reaction (Scheme 1A(d)).[4d] This strategy employed two electrophiles, alkyl bromides and chlorosilanes, to initiate and terminate the reductive cross-coupling, respectively, for accessing silylallenes. Alternatively, decarboxylative silylation of alkyne-functionalized cyclic carbonate derivatives using Cu[4e] or Ni catalysis[4f] for producing various silyl-substituted 2,3- and 3,4-allenols has been developed (Scheme 1A(e)). In contrast, simple, readily accessible propargyl acetates that contain both highly reactive alkyne and acetate moieties can be easily converted into functionalized allenes; however, transition-metal-catalyzed C−Si bond formation for transformations of propargyl acetates to silylallenes remains a formidable challenge due to propargyl acetates having C1 and C3 reactive sites.[5] To address the challenge, we have established an electroreductive cross-coupling method for rendering transformations of prop-2-yn-1-yl acetates with chloro(vinyl)silanes to form tetrasubstituted silylallenes, wherein cathodic reduction of a chlorosilane generates the chlorosilane radical anion and then afford the silyl radical by cleavage of the Si−Cl bond with $Mg^{2+}$ and thus trigger the reductive cross-coupling of prop-2-yn-1-yl acetates

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**Scheme 1. Transition-Metal-Catalyzed Synthesis of Silylallenes**




(Scheme 1A(f)).[4g] However, the reaction requires the use of Mg as the sacrificial electrode and a large amount of $CoCl_2$ as the promoter, thus limiting broad potential applications in synthesis.

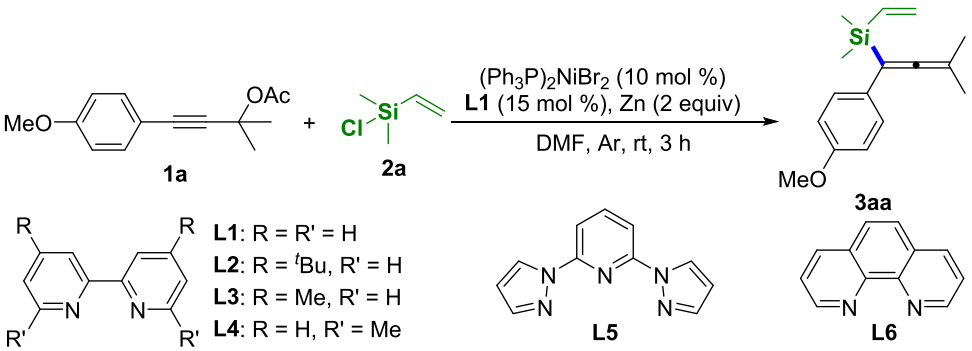
Continuing our research efforts on the reductive cross-coupling,[4g6] we envisioned that using a transition-metal (such as nickel)-catalyzed reductive cross-coupling strategy might initiate oxidative addition of the active Ni(0) species with electrophilic chlorosilanes[7] followed by addition across the alkyne moiety of prop-2-yn-1-yl acetates and deacetoxylation, enabling the synthesis of silylallenes via the C($sp^2$)−Si bond formation. Herein, we report a new, general nickel-catalyzed reductive cross-coupling reaction between propargylic acetates and various substituted chlorosilanes to produce tetrasubstituted silylallenes via the C($sp^2$)−Si bond formation, which is especially dependent on the use of the nitrogen-based ligands (Scheme 1B). We demonstrate that the reaction accommodates to a broad range of substrates under mild conditions and has highly selective catalytic sites by averting the use of a large amount of Co promoter.

## RESULTS AND DISCUSSION

We initiated our investigation by choosing 4-(4-methoxyphenyl)-2-methylbut-3-yn-2-yl acetate (**1a**) and chlorodimethylvinylsilane (**2a**) as the model substrates for the Ni-catalyzed reductive cross-coupling protocol (Table 1). After numerous trials, we found that the combination of $(Ph_3P)_2NiBr_2$ (10 mol %), 2,2′-bipyridine (**L1**) (15 mol %), and Zn (2.0 equiv) in DMF at room temperature for 3 h generated the optimal yield (72%) of the desired (1-(4-methoxyphenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)silane (**3aa**) (entry 1). The $(Ph_3P)_2NiBr_2$, zinc reductant, and ligand **L1** were required for the reaction. If any were absent, the desired product (**3aa**) (entries 2, 9, and 16) was not generated. Although a lower $(Ph_3P)_2NiBr_2$ content (5 mol %) reduced the yield to 67% (entry 3), a higher content (15 mol %) negligibly improved the yield (entry 4), while other nickel catalysts, including


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**Table 1. Screening of the Optimal Reaction Conditions**[a]




| entry | variation from the standard conditions | yield (%) |
|---|---|---|
| 1 | none | 72 |
| 2 | without $(Ph_3P)_2NiBr_2$ | trace |
| 3 | $(Ph_3P)_2NiBr_2$ (5 mol %) | 67 |
| 4 | $(Ph_3P)_2NiBr_2$ (15 mol %) | 70 |
| 5 | $(Ph_3P)_2NiCl_2$ instead of $(Ph_3P)_2NiBr_2$ | 65 |
| 6 | $NiBr_2$·dme instead of $(Ph_3P)_2NiBr_2$ | 15 |
| 7 | $Ni(dppf)Cl_2$ instead of $(Ph_3P)_2NiBr_2$ | 24 |
| 8 | $NiBr_2$ instead of $(Ph_3P)_2NiBr_2$ | 12 |
| 9 | without **L1** | trace |
| 10 | **L1** ( 10 mol %) | 68 |
| 11 | **L2** instead of **L1** | 27 |
| 12 | **L3** instead of **L1** | 32 |
| 13 | **L4** instead of **L1** | 17 |
| 14 | **L5** instead of **L1** | 20 |
| 15 | **L6** instead of **L1** | 50 |
| 16 | without Zn | trace |
| 17 | Mn instead of Zn | 30 |
| 18 | TDAE instead of Zn | trace |
| 19 | $Et_3N$ instead of Zn | trace |
| 20 | DMAc instead of DMF | 40 |
| 21 | NMP instead of DMF | 28 |
| 22 | MeCN instead of DMF | 10 |
| 23 | reaction conducted at 15 °C | 66 |
| 24 | reaction conducted at 35 °C | 61 |
| 25[b] | none | 60 |

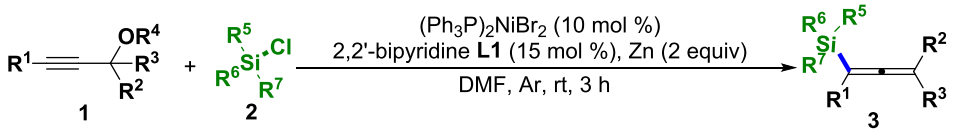
[a]Standard reaction conditions: **1a** (0.2 mmol), **2a** (0.8 mmol), $(Ph_3P)_2NiBr_2$ (10 mol %), **L1** (15 mol %), Zn (2 equiv), DMF (2 mL), argon, room temperature, and 3 h. [b]**1a** (1 mmol), DMF (4 mL), and 6 h.

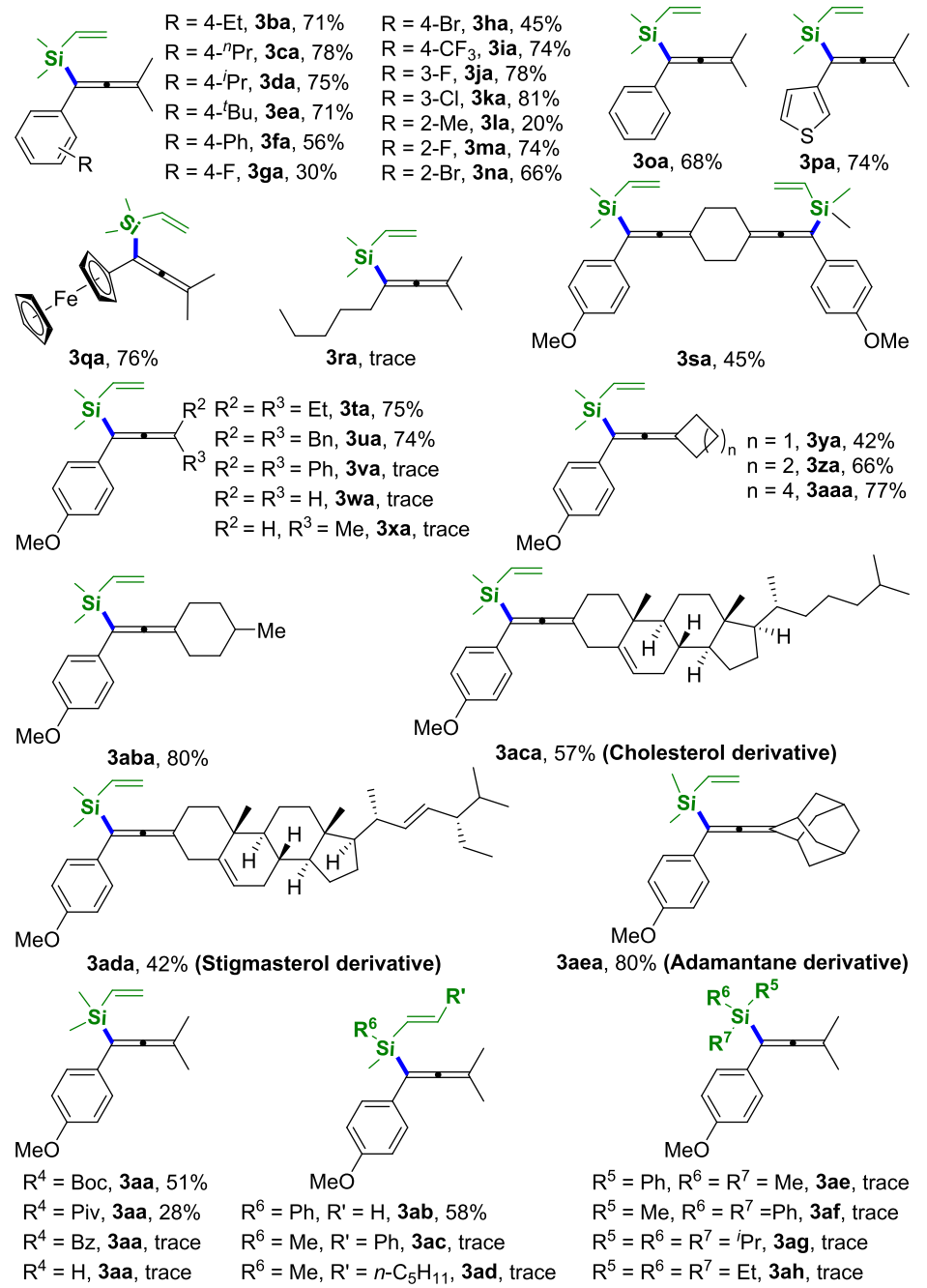
$(Ph_3P)_2NiCl_2$, $NiBr_2$·dme, $Ni(dppf)Cl_2$, and $NiBr_2$, produced lower yields than $(Ph_3P)_2NiBr_2$ (entries 5−8). However, although a lower **L1** content (10 mol %) reduced the yield to 68% (entry 10), other bipyridine ligands, such as **L2**, **L3**, **L4**, and 2,6-di(1-pyrazolyl)pyridine (**L5**), did not produce higher yields than **L1** (entries 11−14); in fact, the yield diminished when 1,10-phenanthroline **L6** (entry 15) was used. The choice of reductant was the key to the reaction: three reductants, including manganese (Mn), *tetrakis*(dimethylamino)ethylene (TDAE), and triethylamine ($Et_3N$), generated lower-than-expected product yields (entries 17−19). When DMF was replaced by other solvents, such as DMAc, NMP, and $CH_3CN$, the product yield decreased (entries 20−22). Briefly filtering the effect of the reaction temperature showed that the reaction at rt optimized the product yield (entries 23 and 24). Then, the reaction was conducted using 1 mmol of 4-(4-methoxyphenyl)-2-methylbut-3-yn-2-yl (1a) under optimized conditions to obtain a 60% yield of **3aa** (entry 25).

Under optimal reaction conditions, the substrate scope of this nickel-catalyzed reductive cross-coupling protocol was investigated (Table 2). Initially, we investigated the applicability of various propargyl acetates possessing an aryl group at the terminal alkyne for the reductive cross-coupling with chlorosilane **2a**, $(Ph_3P)_2NiBr_2$, **L1**, and Zn (**3ba-qa**). A wide range of functionalities, such as Et, $^n$Pr, $^i$Pr, $^t$Bu, Ph, F, Br, and Cl, on the aryl ring at positions *para*, *meta*, or *ortho* were well tolerated, providing **3ba-na** in 20−81% yields. It should be noted that although the steric hindrance and the electron nature of the substitution had no obvious effect on their reactivity, two aryl propargyl acetates **1g** and **1l** bearing a 4-F or a 2-Me group showed rather lower reactivity leading to **3ga** and **3la**, respectively, in 20−30% yields. For 2-methyl-4-phenylbut-3-yn-2-yl acetate **1o**, the reaction executed smoothly and delivered **3oa** in 68% yield. Important functionalities, such as thiophen-3-yl and ferrocene groups, succumbed to the optimal conditions, giving **3pa-qa** in good yields. However, attempts to reductive cross-coupling of propargyl acetate possessing an alkyl group at the terminal alkyne failed (**3ra**). Notably, dual reductive cross-coupling of 1,4-bis-(phenylethynyl)cyclohexane-1,4-diyl diacetate can take place


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Table 2. Variation of the Propargyl Acetates (1) with Chlorosilanes (2)[a]



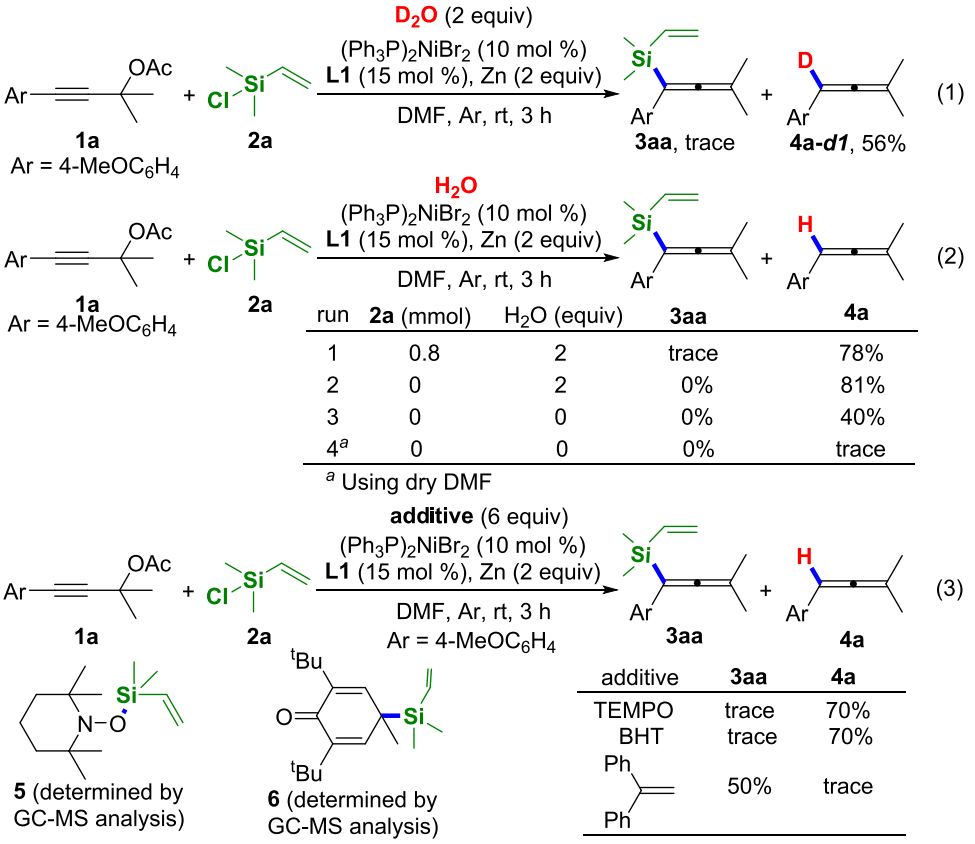




[a]Reaction conditions: **1** (0.2 mmol), **2** (0.8 mmol), $(Ph_3P)_2NiBr_2$ (10 mol %), **L1** (15 mol %), Zn (2 equiv), DMF (2 mL), argon, room temperature, and 3 h.

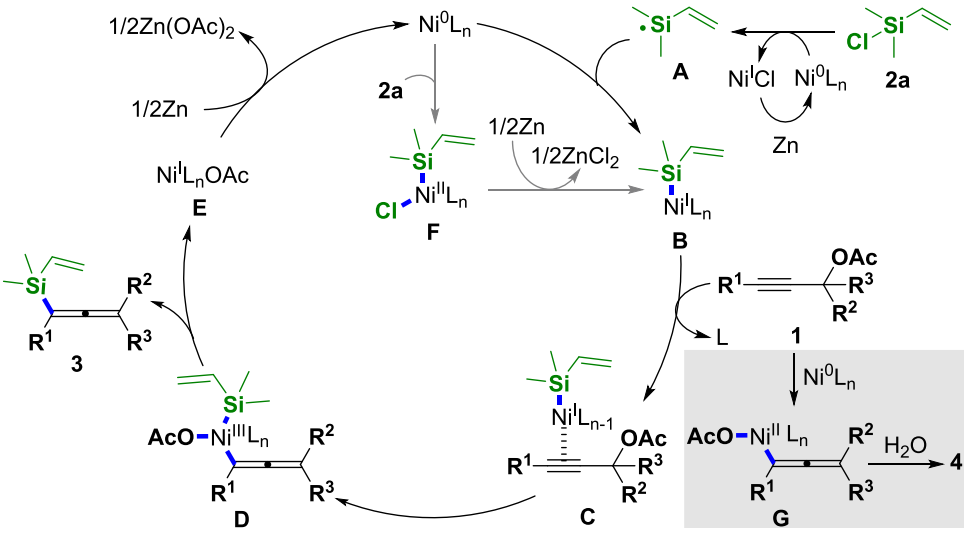
successfully, and one-pot synthesis delivers diallene **3sa**. We found that a wide range of 1,1-dialkyl-substituted propargyl acetates (**1t-u**, **y-ae**) were subject to the optimal conditions, regioselectively generating the corresponding silylallenes **3ta-ua** and **3ya-aea** in moderate to higher yields. However, other propargyl acetates containing 1,1-diaryl, 1,1-nonsubstituted, or 1-monoalkyl, such as 3-(4-methoxyphenyl)-1,1-diphenylprop-2-yn-1-yl acetate (**1v**), 3-(4-methoxyphenyl)prop-2-yn-1-yl acetate (**1w**), and 4-(4-methoxyphenyl)but-3-yn-2-yl acetate (**1x**), had no reactivity for the reductive cross-coupling (**3va-xa**). For example, 1,1-diethyl- or 1,1-dibenzyl-substituted propargyl acetates **1t-u** were converted into **3ta-ua**, respectively, in 75 and 74% yields. Most importantly, propargyl acetates **1y-ae** bearing a cycloalkyl ring at position 1 were


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**Scheme 2. Control Experiments**




**Scheme 3. Proposed Mechanisms**




compatible with the optimal conditions, affording the challenging ethenylidene-cycloalkanes **3ya-aea** in 42−80% yields. Gratifyingly, this reductive cross-coupling protocol can be applicable to late-stage modification of the active pharmaceutical molecules, such as cholesterol derviative,[8] stigmasterol derviative,[8] and adamantane derivative (**3aca**−**3aea**).[8] It was noted that the replacement of OAc by a BocO or a PivO leaving group was reactive to furnish **3aa**, respectively, in 51 and 28% yields, but other leaving groups like BzO and OH instead of OAc had no reactivity.

Next, the scope of chlorosilanes **2** was examined (**3ab-ah**) in the presence of propargyl acetate **1a**, $(Ph_3P)_2NiBr_2$, **L1**, and Zn. Gratifyingly, chloro(methyl)(phenyl)(vinyl)silane **2b** was viable for the construction of **3ab** in 58% yield. However, chlorodimethyl(vinyl)silanes bearing a Ph or an *n*-pentyl group at terminal alkene were inert (**3ac-ad**). Unfortunately, the desired reductive cross-coupling reaction with other nonvinyl-substituted chlorosilanes, such as $Ph(Me)_2SiCl$ **2e**, $Ph_2(Me)$-SiCl **2f**, $^iPr_3SiCl$ **2g**, and $Et_3SiCl$ **2h**, cannot take place (**3ae-ah**). These observations imply that the terminal vinyl moiety is crucial for the success of the reductive cross-coupling protocol,



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probably because the terminal vinyl moiety may exhibit the strong electron and conjugation effect to activate the Si−Cl bond for generating the highly reactive, stable Si-centered radical and initiate the Ni(0)/Ni(I)/Ni(III) catalytic cycle.[7]

To elucidate the mechanism for this nickel-catalyzed reductive cross-coupling protocol, control experiments were conducted (Scheme 2). Under optimal conditions, treatment of propargyl acetate **1a** with chlorosilane **2a** and $D_2O$ afforded the deuterium-labeled allene **4a-*d1***, no target product **3aa**, in 56% yield (eq 1), while the reaction of propargyl acetate **1a** with silane **2a** and $H_2O$ efficiently executed and delivered allene **4a** in 78% yield (run 1; eq 2).[4,5] Notably, without silane **2a**, propargyl acetate **1a** reacted with $H_2O$, $(Ph_3P)_2NiBr_2$, **L1**, and Zn furnished allene **4a** exclusively in 81% yield (run 2; eq 2). While in the absence of both chlorosilanes and $H_2O$, the reaction of propargyl acetate **1a** afforded only **4a** in only 40% yield due to $H_2O$ presented in DMF and the reaction system (run 3; eq 2), no reaction of propargyl acetate **1a** under anhydrous conditions (dry DMF) was observed (run 4; eq 2). These results suggest that in the presence of $H_2O$, the oxidative addition of Ni(0) with propargyl acetate to give the allenylnickel(II) species may take place, which competes with the reductive cross-coupling protocol. When using a radical scavenger, such as 2,2,6,6 tetramethylpiperidin-1-yloxy (TEMPO) and 2,6-di(*tert*-butyl)-4-methylphynol (BHT), the reductive cross-coupling protocol of propargyl acetate **1a** with silane **2a** was completely inhibited and the hydrodeacetoxylation of propargyl acetate **1a** occurred to afford allene **4a** in 70% yield (eq 3). Using 1,1-diphenylethylene as the radical scavenger resulted in decreasing yield of the desired silylallene **3aa** to 50% yield but along with no formation of **4a** (eq 3). These results support that this nickel-catalyzed reductive cross-coupling protocol may involve a free radical process and the oxidative addition of Ni(0) with propargyl acetate to give the allenylnickel(II) species occurs when this nickel-catalyzed radical-mediated reductive cross-coupling protocol is suppressed.

The working mechanisms for this Ni-catalyzed reductive cross-coupling protocol are proposed on the basis of the current results and precedent literature studies (Scheme 3).[4−7] Chlorosilane **2a** may be converted to the Si-centered radical intermediate **A** with the aid of the active $Ni^0L_n$ species and Zn reductant,[7] followed by oxidative addition with the active $Ni^0L_n$ species to form the Si−$Ni^IL_n$ complex intermediate **B**.[4−7] Coordination and decarboxylation of the intermediate **B** affords the Si−$Ni^{III}$(allenyl)-OAc intermediate **D**. Reductive elimination of the intermediate **D** delivers the desired product **3** and the $Ni^IL_n$OAc intermediate **E**. Finally, reduction of the $Ni^IL_n$OAc intermediate **E** by metallic zinc regenerates the active $Ni^0L_n$ species to start a new catalytic cycle. It should be noted that the reductive cross-coupling protocol may proceed via a sequence of direct oxidative addition of chlorosilane **2a** to generate the Si−$Ni^{II}L_n$-Cl intermediate **F** and reduction to form the Si−$Ni^IL_n$ complex intermediate **B**.[4]

In the presence of $H_2O$ or the absence of chlorosilane, the oxidative addition of Ni(0) with propargyl acetate **1** occurs preferentially to generate the allenylnickel(II) species **G**, which would undergo the protonation to afford the allene **4**.

## CONCLUSIONS

In summary, we have developed a novel nickel-catalyzed reductive cross-coupling reaction of propargyl acetates with chloro(vinyl)silanes, enabling the formation of C($sp^2$)−Si bonds to access highly valuable tetrasubstituted silylallenes. The reaction is highlighted by its mild reaction conditions, broad scope of propargyl acetates, good tolerance of functional groups, and excellent selectivity. Most importantly, this reaction can be applicable to late-stage modification of highly valuable natural products and pharmaceuticals.

## EXPERIMENTAL SECTION

**General Considerations.** The $^1H$ and $^{13}C$ NMR spectra were recorded in $CDCl_3$ solvent on an NMR spectrometer using TMS as an internal standard. HRMS was measured on an electrospray ionization (ESI) apparatus using time-of-flight (TOF) mass spectrometry. Melting points were uncorrected. It should be noted that this procedure uses HCONMe2 (DMF) with acute toxicity and must be handled with extreme care.

**Typical Experimental Procedure for Nickel-Catalyzed Reductive Cross-Coupling of Propargylic Acetates (1) with Chloro(vinyl)silanes (2).** To a Schlenk tube were added propargylic acetate **1** (0.2 mmol), chlorosilane **2** (4.0 equiv), $NiBr_2(PPh_3)_2$ (10 mol %), L1 (15 mol %), Zn (2 equiv), and DMF (2 mL). Then, the tube was charged with argon and the mixture was stirred at room temperature for the indicated time (about 3 h) until the complete consumption of the starting material as monitored by TLC analysis. The reaction was quenched with water and extracted with ethyl acetate. The organic layer was washed with saturated $NH_4Cl$ solution, dried over anhydrous $MgSO_4$, filtered, and concentrated in vacuo. The resulting residue was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 100:1 to 50:1 v/v) to afford the desired product **3**.

**Typical Experimental Procedure for a Scale-up to 1 mmol of Propargylic Acetates.** To a Schlenk tube were added 4-(4-methoxyphenyl)-2-methylbut-3-yn-2-yl acetate **1a** (1.0 mmol; 232.1 mg), chlorodimethyl(vinyl)silane **2a** (4.0 equiv, 4.0 mmol; 482.6 mg), $[NiBr_2(PPh_3)_2]$ (10 mol %; 0.1 mmol; 65.4 mg), ligand (15 mol %; 0.15 mmol; 23.4 mg), Zn (2 equiv; 2 mmol; 130.8 mg), and DMF (4 mL). Then, the tube was evacuated briefly under high vacuum and charged with argon using standard Schlenk techniques; this process was repeated three times. The reaction mixture was then stirred at room temperature for the indicated time (about 6 h) until the complete consumption of the starting material, as monitored by TLC analysis. The reaction was quenched with water and extracted with ethyl acetate. The organic layer was washed with saturated $NH_4Cl$ solution, dried over anhydrous $MgSO_4$, filtered, and concentrated in vacuo. The resulting residue was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 100:1, v/v) to afford the desired yellow oil product **3aa** in 60% yield (0.6 mmol; 155.1 mg).

*(1-(4-Methoxyphenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)silane (**3aa**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3aa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 40.3 mg, 78% yield. Colorless oil. $^1H$ NMR (500 MHz, $CDCl_3$): $\delta$ 7.22 (d, $J$ = 9.0 Hz, 2H), 6.81 (d, $J$ = 9.0 Hz, 2H), 6.32−6.23 (m, 1H), 6.02−5.97 (m, 1H), 5.81−5.74 (m, 1H), 3.78 (s, 4H), 1.76 (s, 6H), 0.24 (s, 6H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): $\delta$ 207.9, 157.9, 139.0, 132.3, 132.2, 132.0, 131.9, 130.8, 128.8, 128.5, 128.4, 113.8, 91.6, 55.3, 19.8, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 258 ($M^+$, 19), 243 (85), 173 (20), 115 (9), 85 (100).

*(1-(4-Ethylphenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)silane (**3ba**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ba** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 38.0 mg, 71% yield. Colorless oil. $^1H$ NMR (500 MHz, $CDCl_3$): $\delta$ 7.21 (d, $J$ = 8.0 Hz, 2H), 7.12−7.07 (m, 2H), 6.32−6.23 (m, 1H), 6.02−5.96 (m, 1H), 5.81−5.74 (m, 1H), 2.63−2.57 (m, 2H), 1.75 (s, 6H), 1.21 (t, $J$ = 7.6 Hz, 3H), 0.24 (s, 6H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): $\delta$ 208.3, 141.8, 139.1, 135.7, 132.0, 127.8, 96.7, 28.3, 19.7, 15.54, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 256 ($M^+$, 1), 213 (59), 159 (11), 129 (5), 85 (100).


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*Dimethyl(3-methyl-1-(4-propylphenyl)buta-1,2-dien-1-yl)(vinyl)-silane (**3ca**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ca** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 5:1, v/v). 42.1 mg, 78% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.05 (d, $J$ = 7.5 Hz, 2H), 6.91 (d, $J$ = 7.5 Hz, 2H), 6.17−6.08 (m, 1H), 5.84 (d, $J$ = 14.5 Hz, 1H), 5.62 (d, $J$ = 20.5 Hz, 1H), 2.38 (t, $J$ = 7.5 Hz, 2H), 1.60 (s, 6H), 1.48−1.42 (m, 2H), 0.77 (t, $J$ = 7.0 Hz, 3H), 0.09 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 140.3, 139.1, 135.6, 132.1, 128.4, 127.6, 91.5, 37.7, 24.6, 19.7, 13.9, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 270 ($M^+$, 5), 255 (25), 225 (95), 85 (100), 83 (3). HRMS (ESI) $m/z$: $C_{18}H_{27}Si$ $[M + H]^+$ calcd for 271.1877, found 271.1880.

*(1-(4-Isopropylphenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl-(vinyl)silane (**3da**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3da** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 40.7 mg, 75% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) $\delta$ 7.23 (d, $J$ = 7.5 Hz, 2H), 7.12 (d, $J$ = 7.5 Hz, 2H), 6.33−6.24 (m, 1H), 6.00 (d, $J$ = 14.5 Hz, 1H), 5.78 (d, $J$ = 20.5 Hz, 1H), 2.89−2.82 (m, 1H), 1.75 (s, 6H), 1.23 (d, $J$ = 6.5 Hz, 6H), 0.25 (s, 6H). $^{13}$C NMR ($CDCl_3$, 125 MHz): $\delta$ (ppm) 146.4, 139.1, 135.7, 132.1, 127.7, 126.3, 91.6, 33.7, 24.0, 19.7, −2.0. LRMS (EI, 70 eV) $m/z$ (%): 270 ($M^+$, 1), 255 (8), 227 (100), 185 (15), 85 (97). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{18}H_{27}Si$ 271.1877, found 271.1874.

*(1-(4-(tert-Butyl)phenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl-(vinyl)silane (**3ea**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ea** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 40.4 mg, 71% yield. Colorless oil. $^1$H NMR (400 MHz, $CDCl_3$): $\delta$ 7.30−7.23 (m, 4H), 6.34−6.24 (m, 1H), 6.03−5.96 (m, 1H), 5.82−5.74 (m, 1H), 1.75 (s, 6H), 1.30 (s, 9H), 0.25 (s, 6H). $^{13}$C{$^1$H} NMR (100 MHz, $CDCl_3$): $\delta$ 208.3, 148.7, 139.1, 135.3, 132.1, 127.4, 125.2, 91.6, 34.4, 31.4, 19.7, −2.0. LRMS (EI, 70 eV) $m/z$ (%): 284 ($M^+$, 34), 269 (87), 212 (80), 121 (16), 85 (100). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{19}H_{29}Si$ 285.2033, found 285.2030.

*(1-([1,1′-Biphenyl]-4-yl)-3-methylbuta-1,2-dien-1-yl)dimethyl-(vinyl)silane (**3fa**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3fa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 34.1 mg, 56% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.57 (d, $J$ = 7.5 Hz, 2H), 7.50 (d, $J$ = 7.5 Hz, 2H), 7.43−7.35 (m, 4H), 7.31 (t, $J$ = 7.0 Hz, 1H), 6.35−6.28 (m, 1H), 6.03 (d, $J$ = 14.5 Hz, 1H), 5.81 (d, $J$ = 20.5 Hz, 1H), 1.79 (s, 6H), 0.28 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 141.0, 138.9, 138.7, 137.6, 132.3, 128.7, 128.2, 127.0, 126.9, 91.9, 19.7, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 304 ($M^+$, 19), 289 (61), 232 (28), 202 (16), 85 (100). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{21}H_{25}Si$ 305.1720, found 305.1721.

*(1-(4-Fluorophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (**3ga**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ga** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 14.7 mg, 30% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.31−7.21 (m, 2H), 6.98 (t, $J$ = 8.0 Hz, 2H), 6.34−6.25 (m, 1H), 6.05 (d, $J$ = 14.5 Hz, 1H), 5.81 (d, $J$ = 20.5 Hz, 1H), 1.80 (s, 6H), 0.27 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 208.3, 162.3, 160.3, 138.7, 134.4 (d, $J_{C-F}$ = 3.5 Hz), 132.4, 129.1 (d, $J_{C-F}$ = 8.0 Hz), 96.1, 91.9, 19.7, −2.2. $^{19}$F NMR ($CDCl_3$, 471 MHz) $\delta$ −117.4. LRMS (EI, 70 eV) $m/z$ (%): 246 ($M^+$, 7), 231 (33), 213 (25), 115 (12), 85 (100). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{15}H_{20}FSi$ 247.1313; found 247.1313.

*(1-(4-Bromophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (3ha).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ha** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 27.5 mg, 45% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.36 (d, $J$ = 8.5 Hz, 2H), 7.14 (d, $J$ = 8.5 Hz, 2H), 6.29−6.20 (m, 1H), 6.03−5.98 (m, 1H), 5.80−5.73 (m, 1H), 1.76 (s, 6H), 0.23 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 208.6, 138.6, 137.7, 132.5, 131.3, 129.4, 119.6, 92.2, 19.6, −2.2. LRMS (EI, 70 eV) $m/z$ (%): 306 ($M^+$, 100), 263 (44), 219 (41), 139 (16), 85 (100).

*Dimethyl(3-methyl-1-(4-(trifluoromethyl)phenyl)buta-1,2-dien-1-yl)(vinyl)silane (**3ia**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ia** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 40.4 mg, 74% yield. Colorless oil. $^1$H NMR (400 MHz, $CDCl_3$): $\delta$ 7.35 (d, $J$ = 8.2 Hz, 2H), 7.23 (d, $J$ = 8.2 Hz, 2H), 6.17−6.05 (m, 1H), 5.91−5.84 (m, 1H), 5.68−5.60 (m, 1H), 1.63 (s, 6H), 0.11 (s, 6H). $^{13}$C{$^1$H} NMR (100 MHz, $CDCl_3$): $\delta$ 209.4, 142.7, 138.4, 132.7, 127.9, 126.7 (d, $J$ = 186.4 Hz), 125.12 (m), 96.6, 92.4, 19.5, −2.3. $^{19}$F NMR (376 MHz, $CDCl_3$) $\delta$ −62.3. LRMS (EI, 70 eV) $m/z$ (%): 296 ($M^+$, 5), 227 (44), 192 (66), 85 (100), 83 (2). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{16}H_{20}F_3Si$ 297.1281, found 297.1283.

*(1-(3-Fluorophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (**3ja**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ja** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 38.4 mg, 78% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.04 (q, $J$ = 7.5 Hz, 1H), 6.89 (d, $J$ = 7.5 Hz, 1H), 6.84 (d, $J$ = 10.5 Hz, 1H), 6.72−6.63 (m, 1H), 6.16−6.01 (m, 1H), 5.86 (d, $J$ = 14.5 Hz, 1H), 5.63 (d, $J$ = 20.5 Hz, 1H), 1.61 (s, 6H), 0.09 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 208.7, 163.9, 162.0, 141.1 (d, $J$ = 7.5 Hz), 138.5, 132.5, 129.5 (d, $J$ = 8.5 Hz), 123.4 (d, $J_{C-F}$ = 3.0 Hz), 114.5 (d, $J_{C-F}$ = 21.5 Hz), 112.6 (d, $J_{C-F}$ = 21.5 Hz), 96.5, 92.2, 19.6, −2.2. $^{19}$F NMR ($CDCl_3$, 471 MHz) $\delta$ −113.8. LRMS (EI, 70 eV) $m/z$ (%): 246 ($M^+$, 4), 231 (25), 177 (10), 142 (39), 85 (100). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{15}H_{20}FSi$ 247.1313, found 247.1320.

*(1-(3-Chlorophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (**3ka**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ka** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 42.5 mg, 81% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.11 (s, 1H), 7.05−6.91 (m, 3H), 6.14−6.05 (m, 1H), 5.86 (d, $J$ = 14.5 Hz, 1H), 5.63 (d, $J$ = 20.5 Hz, 1H), 1.61 (s, 6H), 0.08 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 208.7, 140.7, 138.4, 134.1, 132.6, 129.3, 127.8, 125.9, 125.8, 96.3, 92.3, 19.6, −2.2. LRMS (EI, 70 eV) $m/z$ (%): 262 ($M^+$, 2), 247 (16), 227 (42), 85 (100), 83 (2). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{15}H_{20}ClSi$ 263.1017, found 263.1023.

*Dimethyl(3-methyl-1-(o-tolyl)buta-1,2-dien-1-yl)(vinyl)silane (**3la**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3la** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 9.6 mg, 20% yield. Colorless oil. $^1$H NMR (400 MHz, $CDCl_3$): $\delta$ 7.17−7.13 (m, 1H), 7.12−7.06 (m, 2H), 7.05−7.00 (m, 1H), 6.22−6.11 (m, 1H), 5.98−5.92 (m, 1H), 5.74−5.66 (m, 1H), 2.28 (s, 3H), 1.70 (s, 6H), 0.16 (s, 6H). $^{13}$C{$^1$H} NMR (100 MHz, $CDCl_3$): $\delta$ 206.6, 138.5, 138.0, 135.4, 131.9, 130.2, 128.6, 125.7, 125.3, 88.9, 20.8, 19.5, −2.6. LRMS (EI, 70 eV) $m/z$ (%): 242 ($M^+$, 6), 227 (58), 173 (12), 85 (100), 83 (2).

*(1-(2-Fluorophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (**3ma**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ma** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 36.4 mg, 74% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.02 (t, $J$ = 7.5 Hz, 1H), 6.97−6.92 (m, 1H), 6.89−6.80 (m, 2H), 6.10−6.01 (m, 1H), 5.80 (d, $J$ = 14.5 Hz, 1H), 5.56 (d, $J$ = 20.5 Hz, 1H), 1.58 (s, 6H), 0.05 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 209.0, 160.7, 158.7, 138.6 (d, $J_{C-F}$ = 2.5 Hz), 131.8, 130.6 (d, $J_{C-F}$ = 4.0 Hz), 127.4 (d, $J_{C-F}$ = 8.0 Hz), 126.6 (d, $J_{C-F}$ = 15.5 Hz), 123.8 (d, $J_{C-F}$ = 3.5 Hz), 115.4 (d, $J_{C-F}$ = 22.5 Hz), 91.8, 90.7. $^{19}$F NMR ($CDCl_3$, 471 MHz) $\delta$ −113.4. LRMS (EI, 70 eV) $m/z$ (%): 246 ($M^+$, 100), 203 (88), 153 (84), 141 (92), 115 (70). HRMS (ESI) $m/z$: $[M + H]^+$ calcd for $C_{15}H_{20}FSi$ 247.1313, found 247.1316.

*(1-(2-Bromophenyl)-3-methylbuta-1,2-dien-1-yl)dimethyl(vinyl)-silane (**3na**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3na** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 40.4


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mg, 66% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 7.11−7.06 (m, 3H), 6.96 (t, $J$ = 7.0 Hz, 1H), 6.15−6.06 (m, 1H), 5.83 (d, $J$ = 14.5 Hz, 1H), 5.61 (d, $J$ = 20.5 Hz, 1H), 1.59 (s, 6H), 0.07 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 138.9, 138.5, 132.2, 128.2, 127.8, 125.8, 91.6, 19.7, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 306 ($M^+$, 100), 263 (53), 219 (53), 165 (37), 115 (18). HRMS (ESI) $m/z$: [M + H]$^+$ calcd for $C_{15}H_{20}BrSi$ 307.0512, found 307.0503.

*Dimethyl(3-methyl-1-phenylbuta-1,2-dien-1-yl)(vinyl)silane (**3oa**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3oa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 31.0 mg, 68% yield. Colorless oil. $^1$H NMR (400 MHz, $CDCl_3$): 7.31−7.21 (m, 4H), 7.17−7.06 (m, 1H), 6.33−6.23 (m, 1H), 6.03−5.96 (m, 1H), 5.82−5.74 (m, 1H), 1.76 (s, 6H), 0.25 (s, 6H). $^{13}$C{$^1$H} NMR (100 MHz, $CDCl_3$): $\delta$ 208.5, 138.9, 138.6, 132.2, 128.2, 127.8, 125.8, 91.6, 19.7, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 228 ($M^+$, 3), 213 (54), 159 (13), 128 (5), 85 (100).

*Dimethyl(3-methyl-1-(thiophen-3-yl)buta-1,2-dien-1-yl)(vinyl)silane (**3pa**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3pa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 34.6 mg, 74% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.25−7.21 (m, 1H), 7.10 (d, $J$ = 5.0 Hz, 1H), 7.06 (d, $J$ = 1.5 Hz, 1H), 6.37−6.28 (m, 1H), 6.09−6.03 (m, 1H), 5.87−5.80 (m, 1H), 1.79 (s, 6H), 0.29 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 208.1, 138.9, 138.7, 132.4, 128.0, 124.6, 119.8, 91.4, 19.7, −2.4. LRMS (EI, 70 eV) $m/z$ (%): 234 ($M^+$, 19), 219 (40), 144 (20), 129 (10), 85 (100).

*Cyclopenta-1,3-dien-1-yl(4-(1-(dimethyl(vinyl)silyl)-3-methylbuta-1,2-dien-1-yl)cyclopenta-1,3-dien-1-yl)iron (**3qa**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3qa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 100:1, v/v). 51.0 mg, 76% yield. Red oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 6.34−6.25 (m, 1H), 6.02 (d, $J$ = 14.5 Hz, 1H), 5.79 (d, $J$ = 20.5 Hz, 1H), 4.25 (s, 2H), 4.11 (s, 2H), 4.05 (s, 4H), 1.75 (s, 6H), 0.26 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 139.2, 131.8, 90.9, 69.3, 67.7, 67.5, 19.7, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 336 (M+, 100), 321 (27), 251 (27), 121 (30), 85 (22). HRMS (ESI) $m/z$: [M + H]$^+$ calcd for $C_{19}H_{25}FeSi$ 337.1069, found 337.1065.

*(1-(4-Methoxyphenyl)-2-(4-methylenecyclohexylidene)vinyl)-dimethyl(vinyl)silane (**3sa**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3sa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 41.1 mg, 45% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.27−7.19 (m, 4H), 6.84−6.78 (m, 4H), 6.36−6.22 (m, 2H), 6.08−5.95 (m, 2H), 5.83−5.74 (m, 2H), 3.78 (s, 6H), 2.41−2.33 (m, 8H), 0.28 (s, 12H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 204.5, 204.4, 158.1, 138.9, 132.3, 132.3, 130.4, 128.7, 128.7, 113.9, 97.4, 96.8, 55.3, 31.6, 31.5, −2.0, −1.9.

*(3-Ethyl-1-(4-methoxyphenyl)-2l5-penta-1,2-dien-1-yl)dimethyl-(vinyl)silane (**3ta**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ta** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 42.9 mg, 75% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$7.27−7.24 (m, 2H), 6.84−6.77 (m, 2H), 6.33−6.24 (m, 1H), 6.02−5.97 (m, 1H), 5.90−5.64 (m, 1H), 3.78 (s, 3H), 2.11−1.98 (m, 4H), 1.03 (t, $J$ = 7.5 Hz, 6H), 0.24 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 206.3, 157.9, 139.1, 132.1, 128.4, 113.8, 104.8, 55.2, 25.3, 12.5, −2.0.

*(3-Benzyl-1-(4-methoxyphenyl)-4-phenylbuta-1,2-dien-1-yl)-dimethyl(vinyl)silane (**3ua**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ua** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 60.7 mg, 74% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.30 (t, $J$ = 7.5 Hz, 4H), 7.25−7.21 (m, 6H), 7.10 (d, $J$ = 8.5 Hz, 2H), 6.83 (d, $J$ = 9.0 Hz, 2H), 6.19−6.11 (m, 1H), 6.01−5.95 (m, 1H), 5.70−5.63 (m, 1H), 3.82 (s, 3H), 3.45−3.34 (m, 4H), 0.14 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$,): $\delta$ 208.1, 139.5, 138.6, 132.3, 130.2, 129.3, 128.6, 128.1, 126.1, 113.6, 100.8, 55.2, 38.8, −2.3. LRMS (EI, 70 eV) $m/z$ (%): 410 ($M^+$, 32), 319 (100), 165 (26), 91 (91), 85 (99).

*(2-Cyclobutylidene-1-(4-methoxyphenyl)vinyl)dimethyl(vinyl)-silane (**3ya**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ya** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 22.6 mg, 42% yield. Light yellow oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.26 (d, $J$ = 7.5 Hz, 2H), 6.82 (d, $J$ = 8.0 Hz, 2H), 6.34−6.25 (m, 1H), 6.01 (d, $J$ = 14.5 Hz, 1H), 5.78 (d, $J$ = 20.5 Hz, 1H), 3.78 (s, 3H), 2.98 (s, 2H), 2.90 (s, 2H), 1.97 (s, 2H), 0.28 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 201.3, 158.1, 138.9, 132.3, 130.5, 128.8, 113.7, 97.3, 55.2, 29.4, 17.7, −2.1. LRMS (EI, 70 eV) $m/z$ (%): 270 ($M^+$, 23), 242 (56), 227(100), 211 (13), 85 (76). HRMS (ESI) $m/z$: [M + H]$^+$ calcd for $C_{17}H_{23}OSi$ 271.1513,; found 271.1515.

*(2-Cyclopentylidene-1-(4-methoxyphenyl)vinyl)dimethyl(vinyl)-silane (**3za**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3za** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 37.5 mg, 66% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.23 (d, $J$ = 8.5 Hz, 2H), 6.81 (d, $J$ = 8.5 Hz, 2H), 6.33−6.24 (m, 1H), 6.02−5.97 (m, 1H), 5.81−5.74 (m, 1H), 3.78 (s, 4H), 2.56−2.46 (m, 2H), 2.44−2.34 (m, 2H), 1.73−1.67 (m, 4H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 203.5, 157.9, 139.2, 132.1, 130.9, 128.8, 113.8, 113.8, 100.1, 77.2, 55.3, 30.9, 27.4, −2.0. LRMS (EI, 70 eV) $m/z$ (%): 284 ($M^+$, 34), 241 (46), 212 (78), 115 (10), 85 (100).

*(2-Cycloheptylidene-1-(4-methoxyphenyl)vinyl)dimethyl(vinyl)-silane (**3aaa**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3aaa** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 48.0 mg, 77% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.18 (d, $J$ = 8.0 Hz, 1H), 7.13 (d, $J$ = 8.5 Hz, 1H), 6.74 (t, $J$ = 9.0 Hz, 2H), 6.27−6.14 (m, 1H), 5.92 (d, $J$ = 14.5 Hz, 1H), 5.70 (d, $J$ = 20.5 Hz, 1H), 3.70 (s, 3H), 2.23 (s, 2H), 2.08−1.94 (m, 2H), 1.76−1.69 (m, 2H), 1.39 (s, 1H), 1.11−0.98 (m, 2H), 0.85 (t, $J$ = 7.0 Hz, 3H), 0.19 (s, 3H), 0.15 (s, 3H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 204.4, 157.8, 139.1, 132.2, 132.1, 130.7, 128.7, 128.6, 113.8, 113.7, 98.7, 98.1, 55.2, 35.9, 35.7, 32.3, 32.1, 30.5, 30.2, 22.4, 22.2, −1.9, −2.0. LRMS (EI, 70 eV) $m/z$ (%): 312 ($M^+$, 48), 255 (36), 189(26), 121 (17), 85 (100). HRMS (ESI) $m/z$: [M + H]$^+$ calcd for $C_{20}H_{29}OSi$ 313.1982, found 313.1985.

*(1-(4-Methoxyphenyl)-2-(4-methylcyclohexylidene)vinyl)-dimethyl(vinyl)silane (**3aba**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3aba** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 49.9 mg, 80% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.16 (d, $J$ = 8.0 Hz, 2H), 6.74 (d, $J$ = 8.0 Hz, 2H), 6.26−6.17 (m, 1H), 5.93 (d, $J$ = 14.5 Hz, 1H), 5.71 (d, $J$ = 20.5 Hz, 1H), 3.70 (s, 3H), 2.32−2.19 (m, 4H), 1.60 (s, 4H), 1.52 (s, 4H), 0.18 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 208.0, 157.8, 139.2, 132.1, 128.6, 113.7, 100.8, 55.2, 31.9, 29.4, 28.9, −1.9. LRMS (EI, 70 eV) $m/z$ (%): 312 ($M^+$, 36), 241 (42), 145 (32), 121 (22), 85 (100). HRMS (ESI) $m/z$: [M + H]$^+$ calcd for $C_{20}H_{29}OSi$ 313.1982, found 313.1971.

*(2-(10,13-Dimethyl-17-(6-methylheptan-2-yl)-1,2,4,7,8,9,10,11,12,13,14,15,16,17-tetradecahydro-3H-cyclopenta-[a]phenanthren-3-ylidene)-1-(4-methoxyphenyl)vinyl)dimethyl-(vinyl)silane (**3aca**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3aca** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 66.6 mg, 57% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.25−7.21 (m, 2H), 6.82−6.78 (m, 2H), 6.33−6.23 (m, 1H), 6.03−5.96 (m, 1H), 5.81−5.74 (m, 1H), 5.34 (d, $J$ = 2.5 Hz, 1H), 3.77 (s, 3H), 2.66 (d, $J$ = 15.0 Hz, 1H), 2.39−2.19 (m, 2H), 2.06−1.93 (m, 3H), 1.87−1.81 (m, 1H), 1.61−1.45 (m, 5H), 1.41−1.32 (m, 3H), 1.29−1.21 (m, 2H), 1.18−0.98 (m, 12H), 0.94−0.89 (m, 3H), 0.88−0.84 (m, 7H), 0.70 (d, $J$ = 3.0 Hz, 3H), 0.24 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 203.9, 157.9, 139.1, 132.1, 130.7, 128.7, 120.1, 113.8, 98.8, 56.8, 56.2, 55.2, 50.1, 42.3, 40.0, 39.9, 39.5, 37.3, 36.3,


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36.2, 35.8, 31.9, 28.3, 28.0, 26.7, 24.3, 23.9, 22.8, 22.6, 21.0, 18.9, 18.7, 11.9, −1.9, −2.0.

*(2-((8S,9S,10R,13R,14S,17R)-17-((2R,5S,E)-5-ethyl-6-methylhept-3-en-2-yl)-10,13-dimethyl-1,2,4,7,8,9,10,11,12,13,14,15,16,17-tetradecahydro-3H-cyclopenta[a]phenanthren-3-ylidene)-1-(4-methoxyphenyl)-2λ⁵-vinyl)dimethyl(vinyl)silane (stigmasterol derivatives) (**3ada**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3ada** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 51.2 mg, 42% yield. Colorless oil. $^1$H NMR ($CDCl_3$, 500 MHz): $\delta$ (ppm) 1H NMR (500 MHz, Chloroform-*d*) $\delta$ 7.22 (d, *J* = 8.0 Hz, 2H), 6.80 (d, *J* = 7.5 Hz, 2H), 6.31−6.22 (m, 1H), 5.98 (d, *J* = 14.5 Hz, 1H), 5.77 (d, *J* = 20.5 Hz, 1H), 5.34 (s, 1H), 5.23−5.12 (m, 1H), 5.08−4.97 (m, 1H), 3.77 (s, 3H), 3.16 (d, *J* = 14.5 Hz, 1H), 2.63 (d, *J* = 14.5 Hz, 1H), 2.37 (t, *J* = 13.5 Hz, 1H), 2.23 (d, *J* = 13.0 Hz, 1H), 2.07−1.91 (m, 4H), 1.59−1.42 (m, 8H), 1.23 (d, *J* = 28.5 Hz, 6H), 1.06 (s, 4H), 1.04 (s, 5H), 0.85 (s, 3H), 0.81 (s, 6H), 0.71 (s, 3H), 0.23 (s, 6H). $^{13}$C{$^1$H} NMR ($CDCl_3$, 125 MHz): $\delta$ 203.7, 157.9, 139.1, 138.3, 132.1, 130.8, 129.3, 128.7, 120.1, 113.7, 98.4, 97.0, 56.8, 56.0, 55.2, 51.2, 50.4, 42.2, 40.5, 39.9, 39.7, 37.4, 37.1, 31.9, 31.9, 28.9, 27.0, 25.4, 24.4, 21.2, 21.1, 20.96, 19.0, 18.9, 12.3, 12.1, −1.86, −1.9. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{42}H_{63}OSi$ 611.4643, found 611.4647.

*(1-(4-Methoxyphenyl)-2-(octahydro-7H-2,5-methanoinden-7-ylidene)vinyl)dimethyl(vinyl)silane (Adamantane Derivatives) (**3aea**).* Following the typical experimental procedure on a 0.2 mmol scale, compound **3aea** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 56.0 mg, 80% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.20 (d, *J* = 7.0 Hz, 2H), 6.73 (d, *J* = 6.5 Hz, 2H), 6.29−6.16 (m, 1H), 5.92 (d, *J* = 14.5 Hz, 1H), 5.70 (d, *J* = 20.5 Hz, 1H), 3.68 (s, 3H), 2.49 (s, 2H), 1.93−1.80 (m, 10H), 1.74 (s, 2H), 0.20 (s, 6H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 200.9, 157.8, 139.3, 133.8, 133.6, 132.1, 131.1, 128.6, 128.5, 128.5, 128.4, 113.7, 106.2, 55.2, 39.4, 38.4, 37.1, 34.4, 28.2, 28.2, −1.7. LRMS (EI, 70 eV) *m/z* (%): 350 ($M^+$, 90), 335 (50), 265 (43), 189 (100), 85 (78). HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{23}H_{31}OSi$ 351.2139; found 351.2149

*(1-(4-Methoxyphenyl)-3-methylbuta-1,2-dien-1-yl)(methyl)(phenyl)(vinyl)silane (**3ab**).*[4g] Following the typical experimental procedure on a 0.2 mmol scale, compound **3ab** was obtained by silica gel column chromatography (eluent: petroleum ether/EtOAc = 50:1, v/v). 37.2 mg, 58% yield. Colorless oil. $^1$H NMR (500 MHz, $CDCl_3$): $\delta$ 7.57−7.55 (m, 2H), 7.36−7.31 (m, 3H), 7.17 (d, *J* = 8.0 Hz, 2H), 6.75 (d, *J* = 8.0 Hz, 2H), 6.46−6.37 (m, 1H), 6.14−6.08 (m, 1H), 5.84−5.77 (m, 1H), 3.75 (s, 3H), 1.72 (s, 3H), 1.70 (s, 3H), 0.47 (s, 3H). $^{13}$C{$^1$H} NMR (125 MHz, $CDCl_3$): $\delta$ 209.3, 158.0, 137.2, 136.4, 134.5, 134.0, 130.4, 129.1, 129.0, 127.7, 113.8, 94.8, 92.2, 55.2, 19.7, 19.6, −3.3. LRMS (EI, 70 eV) *m/z* (%): 320 ($M^+$, 36), 305 (62), 227 (17), 147 (89), 121 (100).

## ■ ASSOCIATED CONTENT

### Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.3c02228.

Experimental procedure for the starting material (propargylic acetates) synthesis, characterization data for starting materials, copies of $^1$H, and $^{13}$C NMR spectra of all of the synthesized compounds (PDF)

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### Notes

The authors declare no competing financial interest.

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